7. Write a program to implement Logistic Regression (LR) algorithm in python

```
from sklearn.linear_model import LogisticRegression
import matplotlib.pyplot as plt

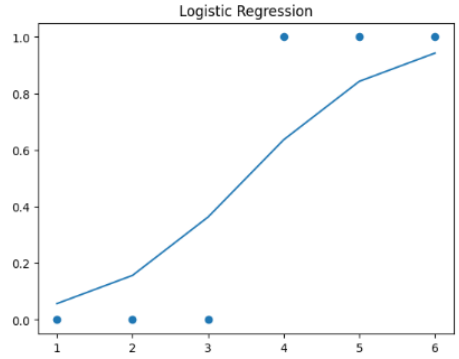
X = [[1],[2],[3],[4],[5],[6]]
y = [0,0,0,1,1,1]

model = LogisticRegression()
model.fit(X,y)

print("Prediction:", model.predict([[5]]))

plt.scatter(X,y)
plt.plot(X,model.predict_proba(X)[:,1])
plt.title("Logistic Regression")
plt.show()
```

```
Prediction: [1]
```



8. Write a program to implement Linear Regression (LR) algorithm in python

```
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt

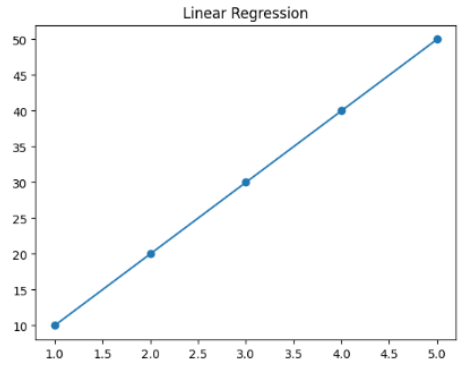
X = [[1],[2],[3],[4],[5]]
y = [10,20,30,40,50]

model = LinearRegression()
model.fit(X,y)

print("Prediction:", model.predict([[6]]))

plt.scatter(X,y)
plt.plot(X,model.predict(X))
plt.title("Linear Regression")
plt.show()
```

```
Prediction: [60.]
```



9. Compare Linear and Polynomial Regression using Python

```
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
import matplotlib.pyplot as plt

X = [[1],[2],[3],[4],[5]]
y = [1,4,9,16,25]

L = LinearRegression()
L.fit(X,y)

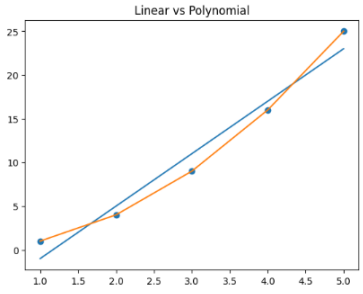
P = PolynomialFeatures(2)
XP = P.fit_transform(X)

PL = LinearRegression()
PL.fit(XP,y)

print("Linear:", L.predict([[6]]))
print("Polynomial:", PL.predict(P.transform([[6]])))

plt.scatter(X,y)
plt.plot(X,L.predict(X))
plt.plot(X,PL.predict(XP))
plt.title("Linear vs Polynomial")
plt.show()
```

```
Linear: [29.]
Polynomial: [36.]
```



10. Write a Python Program to Implement Expectation & Maximization Algorithm

```
from sklearn.mixture import GaussianMixture
import matplotlib.pyplot as plt

X = [[1],[2],[3],[10],[11],[12]]

model = GaussianMixture(2)
model.fit(X)

p = model.predict(X)

print("Clusters:", p)

plt.scatter([1,2,3,10,11,12],
            [0,0,0,0,0,0], c=p)

plt.title("EM Algorithm")
plt.show()
```

```
Clusters: [1 1 1 0 0 0]
```

